

Contents

Awesome Mnemonics - Complete Guide	2
Quick Reference & On-Call Guide	2
Categorized Index	3
Ops	3
RCA (Root Cause Analysis)	3
Systems Design	4
Human Factors	4
Personal Life	4
Who Uses This?	5
Mnemonic Selection Flowchart	5
Table of Contents	6
Problem Solving Techniques	7
Problem Analysis	42
Problem Resolution Threats	54
DICE	54
FATE	54
Communication & Conflict	55
Stress & Resilience	68
PACE	68
STOP	68
ARIES	68
HELP	69
HANDLE	69
CALM	69
PUSH	69
HOPE	69
SHINE	70
Infrastructure & Systems Engineering	70
TRACE (Network Troubleshooting)	70
SCALE (Infrastructure Design)	70
DEBUG (Code & System Analysis)	71
Proven Mnemonic Pipelines	71
1. CRISIS RESPONSE CHAIN	71
2. CONFLICT RESOLUTION CHAIN	72
3. ROOT CAUSE INVESTIGATION CHAIN	72
4. STRATEGIC DESIGN CHAIN	73
5. STRESS BURNOUT RECOVERY CHAIN	74
6. RAPID TRIAGE CHAIN	74
Pipeline Selection Matrix	75
Pro Tips for Using Pipelines	76
Common Mistakes to Avoid	76
Jumping to Solutions	76
Using Wrong Pipeline	76
Incomplete Execution	76

Solo Hero Mode	76
Analysis Paralysis	77
Contributing	77
Sources & References	77
Problem-Solving Methodologies	77
Strategic & Analysis Frameworks	77
Problem Resolution Threats & Triage	78
Communication, Stress & Resilience, Infrastructure	78

Awesome Mnemonics - Complete Guide

Memory hacks that compress complex workflows into actionable acronyms. Use during incidents, RCAs (root cause analysis), design decisions, or high-stress situations.

Disclaimer: This collection includes both established, documented frameworks (e.g. 8D, SWOT, PESTEL, RACI, SET) and mnemonics compiled for learning and incident use. See Sources & References for attribution and classification.

Release v1.5 — 2026-01-25

In PDF and DOCX, the table of contents and in-text links are clickable; use the TOC to jump to sections.

Quick Reference & On-Call Guide

Production incident? → Crisis Response Chain (STOP → TRACE → DEBUG → 8D)

Need quick decision? → Rapid Triage Chain (IDEA → DICE → FATE)

Mobile: Table scrolls horizontally if needed

Situation	Use This	Time
Immediate crisis	IDEA → STOP	2-5 min
Root cause needed	ICEBERG + 5 Whys	30-60 min
Team conflict	BREATHE → PAUSE → WAIT	5-10 min
Strategic planning	PREPARE + SWOT	1-2 hours
System design	SCALE + PESTEL	Planning
Technical debugging	TRACE → DEBUG	Variable
Stress overload	PACE → ARIES → CALM	10-15 min

(If icons don't display in your viewer: =crisis, =root cause, =conflict, =planning, =design, =debug, =stress. Row text is sufficient.)

Quick links: [Index](#) | [Pipelines](#) | [Print](#) | [TOC](#) | [Back to top](#)

Categorized Index

Quick lookup by functional area with “when to use” guidance

Ops

Incident response, troubleshooting, and operational decision-making

Mnemonic	When to Use
STOP	First response to any crisis—immediate stress management (2-5 min)
IDEA	Quick, simple problems requiring fast action (2-5 min)
TRACE	Network/system diagnostics—start with connectivity tests
DEBUG	Code or system analysis when problem definition is unclear
DICE	Identify blockers (Delay, Incompetence, Conflict, External factors) before execution
FATE	Validate resource availability (Funding, Allocation, Time, Expertise) for feasibility
PEST	Identify external threats (Political, Economic, Social, Technological) to solutions

RCA (Root Cause Analysis)

Deep investigation, recurring issues, and prevention

Mnemonic	When to Use
ICEBERG	Complex problems with hidden root causes (30-60 min)
5 Whys	Drill down to root cause—keep asking until systemic failure is found
Ishikawa (Fishbone)	Visual cause-effect analysis; map multiple causes to a problem
8D Approach	Critical incidents requiring formal resolution and prevention
A3 Problem Solving	Structured one-page problem-solving (Toyota); good for sharing and alignment
PADDER	Data-driven problem solving when patterns need identification

Mnemonic	When to Use
PREPARE	Strategic planning for medium complexity problems (1-2 hours)

Systems Design

Architecture, planning, and technical decision-making

Mnemonic	When to Use
SCALE	Infrastructure design—Security, Capacity, Automation, Load balancing, Error handling
SWOT	Evaluate options—Strengths, Weaknesses, Opportunities, Threats (1-2 hours)
PESTEL	Strategic planning considering external factors (Political, Economic, Sociocultural, Technological, Environmental, Legal)
SET	Set stakeholder expectations—pick 2: Good, Fast, Cheap

Human Factors

Team dynamics, conflict resolution, and collaboration

Mnemonic	When to Use
RACI	Resolve role confusion—assign Responsible, Accountable, Consulted, Informed
WAIT	Before speaking—ask “Why am I talking?” to choose the right response
BREATHE	First step when tensions rise—regulate emotions before responding
PAUSE	When BREATHE isn’t enough—step away and evaluate before acting (5-20 min)
DICE	Check for Conflict as a blocker in team execution
FATE	Validate team Expertise and resource Allocation

Personal Life

Stress management, burnout recovery, and personal resilience

Mnemonic	When to Use
PACE	Immediate stress management—Physical activity, Avoid unhealthy behaviors, Coping skills, Emotional awareness
ARIES	Lifestyle changes for long-term stress reduction (2-4 weeks)
CALM	Build resilience—Confidence, Awareness, Logic, Mindfulness
HELP	Stress management strategies—Handle one at a time, Exercise, Learn to relax, Pace yourself
HANDLE	Six stress-management strategies including positive attitude and support system
PUSH	Build positivity—Practice Gratitude, Use affirmations, Spend time with positive people, Have positive outlook
HOPE	Harness positive thoughts, Open up, Prioritize self-care, Exercise regularly
SHINE	Ongoing practice for sustainable positivity—Stay present, Healthy perspective, Identify positive activities, Nourish relationships, Express yourself

Who Uses This?

Teams and engineers using these mnemonics in production:

- *Your team here? Add yourself!*

Using this? Add your organization/team via pull request. See CONTRIBUTING.md for details.

Mnemonic Selection Flowchart

Mobile tip: If the flowchart below is hard to read, use Quick Reference & On-Call Guide or Categorized Index instead.

Problem?

Quick?	Complex?	Stress?
IDEA + STOP	PREPARE/ ICEBERG+ 5 Whys	STOP → PACE

Table of Contents

- Awesome Mnemonics - Complete Guide
 - Quick Reference & On-Call Guide
 - Categorized Index
 - * Ops
 - * RCA (Root Cause Analysis)
 - * Systems Design
 - * Human Factors
 - * Personal Life
 - Who Uses This?
 - Mnemonic Selection Flowchart
 - Table of Contents
 - Problem Solving Techniques
 - * PREPARE
 - * PADDER
 - * ICEBERG
 - * IDEA
 - * 5 Whys
 - * Ishikawa (Fishbone)
 - * 8D Approach
 - * A3 Problem Solving
 - * 5Ps
 - Problem Analysis
 - * RACI
 - * PESTEL
 - * SWOT
 - * SET (Systems Engineering Triangle)
 - Problem Resolution Threats
 - * DICE
 - * FATE
 - * PEST
 - Communication & Conflict
 - * BREATHE
 - * PAUSE
 - * WAIT

- Stress & Resilience
 - * PACE
 - * STOP
 - * ARIES
 - * HELP
 - * HANDLE
 - * CALM
 - * PUSH
 - * HOPE
 - * SHINE
- Infrastructure & Systems Engineering
 - * TRACE (Network Troubleshooting)
 - * SCALE (Infrastructure Design)
 - * DEBUG (Code & System Analysis)
- Proven Mnemonic Pipelines
 - * **1. CRISIS RESPONSE CHAIN**
 - * **2. CONFLICT RESOLUTION CHAIN**
 - * **3. ROOT CAUSE INVESTIGATION CHAIN**
 - * **4. STRATEGIC DESIGN CHAIN**
 - * **5. STRESS BURNOUT RECOVERY CHAIN**
 - * **6. RAPID TRIAGE CHAIN**
- Pipeline Selection Matrix
- Pro Tips for Using Pipelines
- Common Mistakes to Avoid
 - * **Jumping to Solutions**
 - * **Using Wrong Pipeline**
 - * **Incomplete Execution**
 - * **Solo Hero Mode**
 - * **Analysis Paralysis**
- Contributing
- Sources & References

Problem Solving Techniques

Start here for systematic approaches to complex problems

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Problem Analysis

Frameworks for understanding scope and impact

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Problem Resolution Threats

Identify blockers before they derail your solution

DICE

D - Delay
I - Incompetence
C - Conflict
E - External factors

When to use: - Part of IDEA → DICE → FATE rapid triage chain - Identifying blockers before execution - Risk assessment in PREPARE phase

FATE

F - Funding
A - Allocation of resources
T - Time
E - Expertise

When to use: - Resource validation in rapid triage - Feasibility assessment (10-30 min) - After DICE to validate resource availability ### PEST

P - Political
E - Economic
S - Social
T - Technological

When to use: - Identifying external threats to solutions - Consider ways to combat/remove these threats

Communication & Conflict

Stay calm and productive during difficult conversations

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Stress & Resilience

Staying calm and connected during high-pressure work

PACE

- P - Physical activity
- A - Avoiding unhealthy behaviors
- C - Coping skills
- E - Emotional awareness

When to use: - First step in PACE → ARIES → CALM → SHINE burnout recovery chain - Immediate stress management actions

STOP

- S - Step back
- T - Take a deep breath
- O - Observe what is happening
- P - Pull back and put things in perspective

When to use: - First response to ANY crisis - immediate stress management (2-5 min) - Start of STOP → TRACE → DEBUG → 8D crisis response chain - Use with IDEA for quick problem resolution in crisis

Common pitfalls: - **Going through motions** - Rushing through STOP without actually calming. The “Take a deep breath” must be intentional - pause for 3-5 seconds. - **Stopping at STOP** - Using STOP but not proceeding to next step (TRACE/IDEA). STOP is the foundation, not the solution.

Combines well with: BREATHE + PAUSE (breathing techniques), IDEA (quick problem solving)

Real-world example: *Production alert at 2 AM - Step back (don't panic), Take deep breath (reduce adrenaline), Observe (read alert details), Pull back perspective (assess severity before waking team), Then proceed to TRACE for diagnostics*

ARIES

- A - Avoid unnecessary stress
- R - Relax and take breaks
- I - Incorporate physical activity into your routine
- E - Eat a healthy diet
- S - Sleep well

When to use: - Lifestyle changes in PACE → ARIES → CALM chain - Long-term stress reduction (2-4 weeks)

HELP

- This mnemonic can help you remember four ways to manage stress:

H - Handle problems one at a time
E - Exercise regularly
L - Learn to relax
P - Pace yourself and set limits.

HANDLE

- This mnemonic can help you remember six stress-management strategies:

H - Have a positive attitude
A - Avoid unnecessary stress
N - Nurture a strong support system
D - Deal with problems directly
L - Learn to relax
E - Exercise regularly

CALM

C - Confidence: Believe in your abilities and strengths
A - Awareness: Stay conscious of your thoughts and feelings
L - Logic: Use rational thinking to overcome doubts
M - Mindfulness: Practice being present and focused

When to use: - Builds long-term resilience and confidence - Confidence connects to HOPE and SHINE (both build positive self-image) - Use after PACE or ARIES for comprehensive stress management

Combines well with: Positivity (HOPE, SHINE), Stress Management (PACE, ARIES)

PUSH

P - Practice Gratitude
U - Use positive affirmations
S - Spend time with positive people
H - Have a positive outlook

HOPE

H - Harness positive thoughts
O - Open up to others
P - Prioritize self-care
E - Exercise regularly

SHINE

S - Stay present, in the moment
H - Have a healthy positive perspective
I - Identify and do positive activities
N - Nourish positive relationships
E - Express yourself

When to use: - End of PACE → ARIES → CALM → SHINE burnout recovery chain - Ongoing practice for sustainable positivity - Connects to CALM for building positive self-image

Infrastructure & Systems Engineering

Technical mnemonics for infrastructure and DevOps work

TRACE (Network Troubleshooting)

T - Test connectivity (ping, traceroute)
R - Review logs and metrics
A - Analyze packet captures
C - Check configurations
E - Escalate with documented evidence

When to use: - Use with PREPARE and 8D Approach for systematic resolution - Start with T (Test) for quick diagnostics, escalate to A (Analyze) for deep dives - Document findings at each step for 8D Approach's D2 (Describe the problem)

Combines well with: Problem Solving (PREPARE, 8D Approach)

SCALE (Infrastructure Design)

S - Security by design
C - Capacity planning
A - Automation-first
L - Load balancing
E - Error handling/resilience

When to use: - Use SWOT analysis alongside for design validation - Apply PESTEL to evaluate external factors affecting infrastructure - Consider SET Triangle trade-offs (fast/cheap/good) for each component

Combines well with: Problem Analysis (SWOT, PESTEL, SET)

DEBUG (Code & System Analysis)

D - Define the problem (what changed?)
E - Examine error messages/logs
B - Break down into components
U - Understand data flow
G - Generate hypothesis and test

When to use: - Combines 5 Whys questioning with ICEBERG's systematic approach - Start with D (Define) - often the problem is unclear or misstated - Use with TRACE for network/system issues

Combines well with: Problem Solving (5 Whys, ICEBERG), Infrastructure (TRACE)

Proven Mnemonic Pipelines

When certain mnemonics are combined in sequence, they create powerful work-flows that amplify effectiveness. These chains are battle-tested for specific high-pressure scenarios.

1. CRISIS RESPONSE CHAIN

STOP → TRACE → DEBUG → 8D

When to use: Production outages, system failures, critical incidents

The Flow:

S - Step back (STOP)
T - Take a deep breath (STOP)
O - Observe what is happening (STOP)
P - Pull back and put things in perspective (STOP)
↓
T - Test connectivity (TRACE)
R - Review logs and metrics (TRACE)
A - Analyze packet captures (TRACE)
C - Check configurations (TRACE)
E - Escalate with documented evidence (TRACE)
↓
D - Define the problem (DEBUG)
E - Examine error messages/logs (DEBUG)
B - Break down into components (DEBUG)
U - Understand data flow (DEBUG)
G - Generate hypothesis and test (DEBUG)
↓

D1-D8 - Full 8D resolution cycle (8D Approach)

Why it works: STOP stabilizes you emotionally, TRACE gives you diagnostic data, DEBUG structures your analysis, 8D ensures you prevent recurrence.

Time investment: 1-4 hours (depending on complexity)

2. CONFLICT RESOLUTION CHAIN

WAIT → BREATHE → PAUSE → RACI

When to use: Team disagreements, tense meetings, stakeholder conflicts

The Flow:

W - "Why am I talking?" (WAIT - listen first)

↓

B - Breathe deeply and slowly (BREATHE)

R - Remain rational and listen (BREATHE)

E - Empathize with the other person's problem (BREATHE)

A - Ask questions to understand (BREATHE)

↓

P - Put things in perspective (PAUSE)

A - Acknowledge your feelings and theirs (PAUSE)

U - Understand you don't have to react immediately (PAUSE)

S - Step Away from the situation (PAUSE)

E - Evaluate options and plan (PAUSE)

↓

R - Responsible (RACI - clarify roles)

A - Accountable (RACI)

C - Consulted (RACI)

I - Informed (RACI)

Why it works: WAIT prevents escalation, BREATHE regulates emotions, PAUSE creates space for rational thought, RACI resolves role confusion (often the root cause).

Time investment: 5-20 minutes

3. ROOT CAUSE INVESTIGATION CHAIN

ICEBERG → 5 Whys → PADDER → RACI

When to use: Recurring issues, complex system problems, post-incident analysis

The Flow:

I - Identify issue(s) (ICEBERG)
 C - Collect data and analyze situation (ICEBERG)
 E - Examine possible root causes (ICEBERG)
 ↓
 Why #1 → Why #2 → Why #3 → Why #4 → Why #5 (5 Whys)
 ↓
 D - Develop solution (PADDER)
 D - Design action plan (PADDER)
 E - Execute action plan & Monitor (PADDER)
 R - Reevaluate and refine (PADDER)
 ↓
 R/A/C/I - Assign responsibilities (RACI)

Why it works: ICEBERG provides structure, 5 Whys drills to root cause, PADDER creates action plan, RACI ensures accountability.

Time investment: 1-2 hours

4. STRATEGIC DESIGN CHAIN

SCALE → SWOT → PESTEL → SET

When to use: Infrastructure planning, architecture reviews, capacity planning

The Flow:

S - Security by design (SCALE)
 C - Capacity planning (SCALE)
 A - Automation-first (SCALE)
 L - Load balancing (SCALE)
 E - Error handling/resilience (SCALE)
 ↓
 S - Strengths (SWOT - evaluate design)
 W - Weaknesses (SWOT)
 O - Opportunities (SWOT)
 T - Threats (SWOT)
 ↓
 P - Political (PESTEL - external factors)
 E - Economic (PESTEL)
 S - Sociocultural (PESTEL)
 T - Technological (PESTEL)
 E - Environmental (PESTEL)
 L - Legal (PESTEL)
 ↓
 Pick 2: Good / Fast / Cheap (SET Triangle - set expectations)

Why it works: SCALE sets technical requirements, SWOT evaluates ap-

proach, PESTEL identifies external risks, SET manages stakeholder expectations.

Time investment: 2-4 hours (planning phase)

5. STRESS BURNOUT RECOVERY CHAIN

PACE → ARIES → CALM → SHINE

When to use: Long-term stress, approaching burnout, need for lifestyle reset

The Flow:

P - Physical activity (PACE - immediate actions)
A - Avoiding unhealthy behaviors (PACE)
C - Coping skills (PACE)
E - Emotional awareness (PACE)
↓
A - Avoid unnecessary stress (ARIES - lifestyle changes)
R - Relax and take breaks (ARIES)
I - Incorporate physical activity (ARIES)
E - Eat a healthy diet (ARIES)
S - Sleep well (ARIES)
↓
C - Confidence (CALM - mental framework)
A - Awareness (CALM)
L - Logic (CALM)
M - Mindfulness (CALM)
↓
S - Stay present (SHINE - ongoing practice)
H - Have healthy perspective (SHINE)
I - Identify positive activities (SHINE)
N - Nourish relationships (SHINE)
E - Express yourself (SHINE)

Why it works: PACE handles immediate stress, ARIES addresses root lifestyle causes, CALM builds mental resilience, SHINE creates sustainable positivity.

Time investment: Ongoing (2-4 weeks for habit formation)

6. RAPID TRIAGE CHAIN

IDEA → DICE → FATE

When to use: Quick wins needed, time-critical decisions, assessing feasibility

The Flow:

I - Identify problem (IDEA)
D - Develop Solution (IDEA)
E - Execute Solution (IDEA)
A - Assess Solution (IDEA)

↓

D - Delay (DICE - check for blockers)
I - Incompetence (DICE)
C - Conflict (DICE)
E - External factors (DICE)

↓

F - Funding (FATE - resource check)
A - Allocation of resources (FATE)
T - Time (FATE)
E - Expertise (FATE)

Why it works: IDEA provides fast framework, DICE identifies blockers, FATE validates resource availability.

Time investment: 10-30 minutes

Pipeline Selection Matrix

Your Situation	Pipeline	Key Benefit
System down NOW	STOP → TRACE → DEBUG → 8D	Systematic crisis response
Team conflict escalating	WAIT → BREATHE → PAUSE → RACI	De-escalation + role clarity
Same issue keeps happening	ICEBERG → 5 Whys → PADDER → RACI	Deep root cause + prevention
Designing new infrastructure	SCALE → SWOT → PESTEL → SET	Complete planning framework
Feeling burned out	PACE → ARIES → CALM → SHINE	Comprehensive stress recovery
Need quick decision	IDEA → DICE → FATE	Fast feasibility assessment

Pro Tips for Using Pipelines

1. **Don't skip steps** - Each mnemonic builds on the previous one
 2. **Document at each stage** - Your STOP insights inform TRACE, TRACE data feeds DEBUG, etc.
 3. **Know when to branch** - If IDEA reveals complexity, switch to ICEBERG → 5 Whys
 4. **Combine with stress tools** - Always start with STOP if you're emotionally activated
 5. **Teach your team** - Shared vocabulary accelerates collaboration
-

Common Mistakes to Avoid

Jumping to Solutions

Problem: Skipping STOP or WAIT when stressed → poor decisions

Fix: Always use STOP first in crises, WAIT before reacting in conflicts

Example: *Production alert at 2 AM - Jumping straight to SSH without STOP → panic-driven wrong server reboot → extended downtime. Instead: STOP (30 seconds) → assess severity → then TRACE*

Using Wrong Pipeline

Problem: IDEA won't solve systemic issues

Fix: Use ICEBERG → 5 Whys for recurring problems

Example: *Database timeouts happen weekly - Using IDEA to restart service each time → problem returns. Instead: ICEBERG → 5 Whys reveals missing connection pool config → permanent fix*

Incomplete Execution

Problem: Starting 8D Approach but stopping at D3 (band-aid)

Fix: Always reach D7 (Prevent Reoccurrence) or the problem returns

Solo Hero Mode

Problem: Forgetting RACI → no accountability when you're unavailable

Fix: Use RACI in 8D Approach's D1 (Form a team)

Analysis Paralysis

Problem: PREPARE → ICEBERG → 5 Whys → 8D for simple problems

Fix: Start with IDEA, escalate only if complexity emerges

[↑ Back to Quick Reference & On-Call Guide](#) | [↑ Back to Top](#)

Contributing

Got a mnemonic that's saved you countless times? **Share it!**

Submission criteria: - Must be provable/actionable (not just motivational)
- Should cross-reference existing mnemonics where applicable - Include real-world usage context - Keep it memorable (that's the point!)

See CONTRIBUTING.md for details.

Sources & References

Problem-Solving Methodologies

- **8D (Eight Disciplines)** — Ford Motor Company (1987). *Team Oriented Problem Solving Manual*. Evolved from TQM; in wide use in automotive and aerospace. Wikipedia
- **5 Whys** — Toyota Production System root cause technique; widely adapted across industries.
- **Ishikawa (Fishbone) Diagram** — Ishikawa, K. (1960s). University of Tokyo. Cause-effect diagram; central to Japanese quality control and Toyota. Wikipedia
- **A3 Problem Solving** — Toyota Production System. Single A3-page structured problem-solving (plan-do-check-act); one-page report for alignment. Wikipedia
- **PADDER, ICEBERG, IDEA, PREPARE** — Curated/educational problem-solving mnemonics for this collection.

Strategic & Analysis Frameworks

- **PESTEL** — Aguilar, F. (1967). *Scanning the Business Environment*. Harvard; later extended to PESTLE/PESTEL (Legal, Environmental). Background
- **PEST** — Four-factor variant (Political, Economic, Social, Technological); conceptually from the PESTEL lineage.

- **SWOT** — Humphrey, A. (1960s–70s). Stanford Research Institute (SRI International); developed with Fortune 500 planning research. e.g. Ninety
- **RACI** — Responsibility Assignment Matrix; emerged ~1950s–70s, no single inventor. Wikipedia RAM
- **SET (Systems Engineering Triangle)** — SEBoK, *Guide to the Systems Engineering Body of Knowledge*. SEBoK

Problem Resolution Threats & Triage

- **DICE, FATE** — Educational mnemonics for blockers and resource checks in this collection.

Communication, Stress & Resilience, Infrastructure

- **BREATHE, PAUSE, WAIT, PACE, STOP, ARIES, HELP, HANDLE, CALM, PUSH, HOPE, SHINE** — Curated for stress management and conflict de-escalation.
- **TRACE, SCALE, DEBUG** — Curated for infrastructure and troubleshooting.

Note: = documented origin; = widely used, adapted from a known source; = compiled/curated for learning in this guide. Some entries are established frameworks; others are educational compilations.